One reason could be that the perturbation in the least-likely attacks is larger and the denoising techniques may work more effectively. In comparison, removing the carefully injected tiny noise in the most-likely attacks may become harder. Finally, we observe that the input denoising ensemble (see last row of each dataset) show stronger defense compared to individual technique alone, while offering competitive benign test accuracy and providing significantly improved robustness under all 11 attacks. Figure 3 shows an intuitive illustration for CIFAR-10.

3.3 Output Verification Ensemble

Ensemble Diversity. We define ensemble diversity in terms of both model structure-based diversity and model disagreement-based diversity [41]. We say that an ensemble meets the structural diversity if the member models of an ensemble are trained using different training process, such as varying training dataset, initial weight filters, neural network structure, neural network algorithm employed (e.g., LeNet, VGG, MobileNet, ResNet-50, ResNet-101, ResNet-152), or by using different settings of hyperparameters (e.g., feature vector size, mini-batch size, # epochs, # iterations, learning rate functions, optimization algorithms). Thus, the structural diversity of ensemble is related to the diversity inherent in the algorithms and the hyperparameter settings used for the DNN model training. The disagreement diversity of an ensemble is related to the outputs of the DNN ensemble, which promotes the failure independence of ensemble member models and increases the overall predictive performance (accuracy).

Several popular diversity metrics, e.g., kappa statistics [37], Q statistic [42], correlation coefficient, fail/non-fail disagreement measure [43]), double-fault measure [44] can be used for selecting most disagreement-diverse models from the N_{out} baseline candidate models trained over the same benign training set. In the first prototype of XEnsemble, we use the Kappa κ statistics, a pairwise metric, to measure the diversity degree between two models using negative examples. Let N denote the number of prediction results, K denote the number of classes, N_{ij} denote the number of instances labeled as class i by one model and as class j by the other model. The κ metric is defined by

$$f_\kappa = \frac{\frac{\sum_{i=1}^{K} N_{ii}}{N} - \sum_{i=1}^{K} \left(\frac{N_{i*}}{N} - \frac{N_{*i}}{N} \right)}{1 - \sum_{i=1}^{K} \left(\frac{N_{i*}}{N} - \frac{N_{*i}}{N} \right)} \quad (1)$$

In Equation 1, $\frac{\sum_{i=1}^{K} N_{ii}}{N}$ computes the percentage of agreement made by the two classifiers i and j under the same series of queries. $\sum_{i=1}^{K} (\frac{N_{i*}}{N} - \frac{N_{*i}}{N})$ computes the chance of agreement in which the $*$ is any label in the output space. κ value is closer to 1 when two models have more agreements, and is closer to 0 if there is more discrepancy. The disagreement diversity of an ensemble team of size M is computed by averaging all pairwise κ values of its member models.

Creating Diversity Ensemble. We develop a two-step diversity ensemble creation algorithm: First, we create a pool of candidate ensemble member models, or so-called base models. Each base model should meet the following two criteria: (1) the structure-based ensemble diversity, and (2) the high benign test accuracy comparable to that of the target model. For benchmark datasets like CIFAR-10, CIFAR-100, ImageNet-1000, instead of training N redundant DNN models, one can also collect those pre-trained DNN models on these benchmark datasets from the public domain. Although generating a pool of base models for datasets in some application domain can be potentially expensive, there are several recent efforts [45], [46] present encouraging mechanisms to create multiple base models conveniently, such as within a single training phase using Snapshots. Second, we need to create disagreement-based ensemble teams from the base model pool. This can be carried out by finding all the subsets of the base model pool of size P, which meet the following two constraints: (i) we want to include the target model in every defense ensemble; and (ii) we want to remove all ensemble teams of size 2. Thus, instead of considering 2^P subsets, we only need to consider 2^{P-2} subsets to create the pool of disagreement-based diversity ensemble teams from a base model pool of size P. We refer to those subsets as the ensemble candidate teams. For each candidate teams, we compute its ensemble disagreement score based on the chosen diversity metric, such as the Kappa κ value. Then we rank all 2^{P-2} candidate ensemble teams by their κ scores and form a pool of κ ensembles by selecting the top-ranked ensemble teams whose κ values are lower than the system-defined κ diversity threshold. The lower the κ value of an ensemble, the higher the disagreement-based ensemble diversity.

Defense with Output Verification Ensemble. The output verification ensemble defense involves two problems: (1) We need to develop robust ensemble consensus methods, which can effectively combine, rank, and integrate predictions from members of an ensemble committee to produce the ensemble prediction output with high accuracy, aiming to improve the robustness of the target model under protection against in adversarial examples and out-of-distribution examples. (2) At prediction phase, for each query input, one ensemble team should be chosen randomly from the disagreement diversity ranked pool of ensemble teams as the chosen output verification team to verify and repair the prediction of the target model by simply providing the ensemble recommendation as the verified output of the target model.

There are several consensus methods for combining the outputs of multiple models in an ensemble of size M. For example, an ensemble output can be created by aggregating the outputs from its M member DNN models via simple averaging (or sum, max, median, min) or a weighted averaging method. The weighted averaging embraces the relative accuracy of the ensemble member DNN classifiers, e.g., the confidence of the top-1 class label. In general, averaging and weighted averaging are popular aggregation methods for the linear opinion pools. Voting is a representative non-linear combining method, which combines the individual votes from M member models using rank-based information. The majority voting is the simplest method, which chooses the classification made by more than half of the DNN member classifiers. When there is no agreement among more than 50% of the DNN member models, the ensemble result is considered an error. The downside of the majority is the scenario where 50% of an ensemble committee misclassify, and a majority voting, in this case, results in ensemble error. The plurality voting method improves the majority in the sense that the collective decision is the classification reached by more DNN classifiers than any other. A correct decision by the majority is inevitably a correct decision by plurality, but not vice versa [41].

We illustrate the model verification ensemble defense against adversarial examples and out-of-distribution attacks in Figure 4. To show the ensemble with only structure diversity and the ensemble with both structure and disagreement diversity, we verify the prediction of the target model using five individual DNN verification models as the pool of base models. We construct

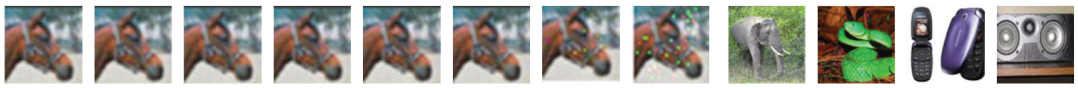
	benign	FGSM	BIM	PGD	CW_∞ LL	CW_2 LL	CW_0 LL	JSMA LL	OOD			
Adversarial Attack image												
target model	horse	bird	bird	bird	bird	airplane	airplane	airplane	cat	frog	airplane	dog
verification model 1	horse	horse	bird	horse	horse	horse	horse	airplane	dog	cat	automobile	horse
verification model 2	horse	bird	horse	bird	bird	cat	bird	horse	ship	deer	dog	truck
verification model 3	horse	horse	deer	horse	deer	airplane	horse	horse	frog	truck	truck	cat
verification model 4	horse	horse	frog	horse	horse	horse	horse	horse	cat	deer	dog	airplane
verification model 5	horse	deer	deer	horse	horse	cat	horse	bird	airplane	ship	automobile	deer
random (VM 2,4,5)	horse	flag	flag	horse	horse	cat	horse	horse	flag	deer	dog	flag
κ selected (VM 1,3,4)	horse	horse	flag	horse	horse	horse	horse	horse	flag	flag	flag	flag

Fig. 4: An illustration of XEnsemble: model verification ensemble defense against adversarial examples and out-of-distribution attacks.

CIFAR-10	κ degree of disagreement					
	DenseNet	ResNet-20	ResNet-32	ResNet-44	ResNet-56	ResNet-110
TM: DenseNet	1	0.382	0.405	0.408	0.44	0.425
VM1: ResNet-20		1	0.496	0.435	0.452	0.478
VM2: ResNet-32			1	0.46	0.469	0.507
VM3: ResNet-44				1	0.464	0.481
VM4: ResNet-56						0.49
VM5: ResNet-110						1
benign acc.	0.9484	0.9184	0.9228	0.9238	0.9277	0.9262

(a) Kappa statistic for CIFAR-10 models

ImageNet	κ degree of disagreement				
	MobileNet	VGG-16	VGG-19	ResNet-50	Inception-v3
TM: MobileNet	1	0.238	0.318	0.307	0.365
VM1: VGG16		1	0.456	0.341	0.284
VM2: VGG19			1	0.33	0.285
VM3: ResNet-50				1	0.399
VM4: Inception-v3					1
benign acc.	0.695	0.68	0.67	0.73	0.67

(b) Kappa statistic for ImageNet models

TABLE 3: Baseline verification model pools for CIFAR-10 and ImageNet with benign test accuracy and pairwise κ measures.

ensemble teams of structure diversity by simply choosing a subset of size 3 or larger from the base model pool, which we refer to as the random ensemble team. We also construct a pool of disagreement diversity ensemble teams, which we refer to as the Kappa κ ensemble. We highlight those examples that are correctly repaired in green by the random ensemble of a selection of base models and by the Kappa κ ranked ensemble team and highlight those that are detected as deceptive inputs in red, which are either OOD inputs or adversarial examples that cannot be repaired by the chosen ensemble team. We observe that the random ensemble is less diverse compared to the Kappa κ ranked ensemble team. For instance, two verifiers out of the random ensemble of three base models (Random (VMs 2,4,5) have made misclassification consistently into the same wrong class (VM2 and VM5), e.g., the adversarial input under CW_2 attack is misclassified into cat, and the two OOD inputs (Frog and cellphone) are misclassified into deer and dog respectively. This comparison also shows the importance of developing a comprehensive ensemble ranking algorithm that takes into account not only the prediction outcome but also other factors, such as the prediction confidence, the L_1 distance of the verification vector to the target prediction vector. A preliminary version of XEnsemble on the robustness of adversarial example is provided in our conference version [47].

4 Experimental Evaluation

4.1 Output Verification Implementation

Datasets. Two popular benchmark datasets CIFAR-10 and ImageNet are used for all experiments reported in this paper. The CIFAR-10 dataset consists of 60,000 color images of objects with a size of 32*32 in 10 classes. 50,000 of them are training data and the rest 10,000 are test data. The ImageNet dataset is provided by the ImageNet Large Scale Visual Recognition Competition (ILSVRC) as a large-scale color image classification benchmark. The dataset contains 1.2 million training images and 50,000 validation images in 1000 classes with a size of 224*224. We also use the TinyImageNet data and LSUN data as the two out-of-distribution datasets, which are fed into a DenseNet classifier trained on CIFAR-10 and CIFAR-100 for the effectiveness evaluation and comparative analysis. CIFAR-100 dataset consists of 50,000 32x32 color training images, labeled with 100 classes, and 10,000 test images. TinyImageNet is a subset of the ImageNet dataset, containing 10,000 test images from 200 different classes. The Large-scale Scene UNderstanding dataset (LSUN) has a test set of 10,000 images on 10 different scenes.

Creating Base Model Pool. For CIFAR-10 trained on DenseNet with test accuracy of 0.9184, we create a base model pool of 5 pre-trained models as candidate verifiers for the target DenseNet model: ResNet-20, ResNet-32, ResNet-44, ResNet-56, and ResNet-110, all obtained from Keras. For ImageNet trained on MobileNet with test accuracy of 0.67, we utilize 4 pre-trained models from Keras as the verification models (VMs) for the target MobileNet model: VGG-16, VGG-19, ResNet-50, and Inception-V3. In addition to the structure-based ensemble diversity, we also use the benign test accuracy as another criterion for our base model selection decision, which requires each model verifier is trained on the same dataset with the same learning task and has the benign test accuracy comparable to that of the target model. Table 3 shows that all base verification models for CIFAR-10 models have a benign accuracy higher than 0.9184 and all ImageNet baseline verification models have a benign accuracy higher than 0.67.

Creating Diversity Ensemble Pool. Next, we create a pool of disagreement-based diversity ensembles by pruning those ensembles with low disagreement diversity. In the first prototype of XEnsemble, we quantify the ensemble disagreement diversity using Kappa(κ) measure. Given that Kappa(κ) measure is a pairwise method, we compute the Kappa(κ) score for a given ensemble of size M by computing the average Kappa score, denoted by κ-average, over all pairwise Kappa(κ) scores of an ensemble. We identify the best Kappa(κ) ensemble team by selecting the top-1

CIFAR-10					ImageNet			
#model	2	3	4	5	#model	2	3	4
top 1	1,3	1,3,4	1,3,4,5	1-5	top 1	1,3	1,3,4	1-4
κ	0.408	0.43	0.441	0.453	κ	0.295	0.322	0.332
acc	0.9476	0.9604	0.9365	0.95	acc	0.775	0.755	0.805
top 2	2,3	1,2,3	1,3,4,5	-	top 2	1,4	1,2,4	-
κ	0.424	0.431	0.446	-	κ	0.296	0.324	-
acc	0.9285	0.9324	0.9301	-	acc	0.77	0.75	-
top 3	1,4	1,3,5	1,2,3,5	-	top 3	2,3	1,2,3	-
κ	0.428	0.435	0.448	-	κ	0.318	0.332	-
acc	0.9198	0.8928	0.9256	-	acc	0.75	0.745	-

TABLE 4: κ value and the benign ensemble accuracy of Top 3 ranked κ ensemble teams of different sizes for CIFAR-10 and ImageNet,

ranked ensemble in the pool of disagreement ensemble. There are different ways to create a pool of disagreement based verification ensembles. Alternative to the approach discussed in Section 3.2, we can build this pool by creating multiple κ ranked ensemble list, one for each size of the ensembles such as 3, 4, 5, 6 for the target DenseNet model (TM) on CIFAR-10 and the base model pool of 5 different Resnet model verifiers. For each Kappa (κ) ranked ensemble list of a given size, we add the verification ensemble with the smallest average pairwise Kappa (κ) value to the disagreement based ensemble pool. Table 4 shows the pool of top-3 Kappa (κ) ensembles with ensemble prediction accuracy and average pairwise κ value for both DenseNet on CIFAR-10 and MobileNet on ImageNet-1000.

Ensemble Ranking. There are several ways to perform prediction ensemble ranking. In our first prototype implementation, we choose to use the L_1 distance and the prediction confidence as two metrics to design our ranking algorithm. We choose the L_1 distance because we only consider those verification models with high benign test accuracy. Then, the predictions for a benign example on both the target model and verification models should be alike. Thus, the L_1 distance of their prediction vectors should be small and close to zero. At the same time, the predictions for an adversarial example are likely to be quite different on the target model and verification model. This is feasible since the transferability on targeted attacks is weak. In the meantime, transferability on untargeted attacks merely causes misclassification and the misclassified destination different across prediction models. Therefore, the L_1 distance between prediction made by the target model and by the verification is relatively larger (higher than 1 and close to 2). Consider k verification versions to the target model (TM), we produce a L_1 ranking of all k prediction outcomes. If the top-ranked recommendations have approximately the same L_1 ranking score, we use the prediction confidence for each of the prediction recommendations to break the tie.

4.2 Defense Evaluation Metrics

We evaluate the effectiveness of our model verification ensemble defense approach and compare it with other existing defenses by a number of cost and effect metrics.

Defense Success Rate (DSR). DSR measures the success rate of the defense system by verifying, repairing and flagging adversarial examples and out-of-distribution examples.

Prevention Success Rate (PSR). PSR measures the percentage of adversarial examples and out-of-distribution examples that are repaired and correctly classified by the defense.

Detection Success Rate (TSR). TSR measures the percentage of adversarial examples and out-of-distribution examples that cannot be repaired but are flagged by the defense.

When adversarial examples and out-of-distribution examples are presented as threats to a target DNN model, the defense success rate (DSR) is the sum of PSR and TSR, i.e., DSR = PSR + TSR. Since there is no correct label for the out-of-distribution input in a trained model, the PSR for OOD is 0 and DSR equals to TSR. To better measure the detection performance of XEnsemble, we also use the following common metrics to evaluate the effectiveness of model verification ensemble defense for detecting out-of-distribution examples in addition to TSR.

Detection error (DError) measures the error made both on out-of-distribution and in-distribution data. It is computed by β(1-TPR)+(1-β)FPR. TPR is computed by TP / (TP+FN), where TP denotes the portion of correctly detected out-of-distribution examples and FN denotes the portion of OOD inputs that are considered as in-distribution data. FPR is computed by FP/(FP+TN), where FP is the proportion of in-distribution inputs that are identified as out-of-distribution inputs and TN is the proportion of in-distribution data that are identified correctly. We set $\beta = 0.5$.

AUROC is a threshold-independent metric that measures the tradeoff between (1-FPR) and TPR. The range of AUROC is [0,1]. It can be interpreted as the probability that a positive example is assigned a higher detection score than a negative example. The better the detection performance is, the larger the AUROC will be.

4.3 Defending Adversarial Attacks on CIFAR-10

We evaluate the effectiveness of XEnsemble defense for verifying and repairing the prediction decision made by the target model in the presence of adversarial examples. This set of experiments compares three different model verification ensemble defense options: XEnsemble-rand, XEnsemble-κ-rand, and XEnsemble-best-κ. XEnsemble-rand is an ensemble created from the base model pool with structure-based diversity. XEnsemble-κ-rand is an ensemble with disagreement diversity, selected randomly from the Kappa ranked list of ensembles whose Kappa ensemble diversity measure is above the system-defined diversity threshold. XEnsemble-best-κ is the ensemble with the highest diversity in the Kappa κ ranked list of ensembles. For CIFAR-10, the XEnsemble-best-κ ensemble is VM 1,3,4. We provide the defense performance measurement in Table 5. We observe three interesting facts: (1) XEnsemble strategy is always beneficial regardless whether it is the random ensemble from the baseline verification model pool, or the random κ ensemble from the pool of κ ranked ensemble list, or the best κ ensemble. (2) the best κ model ensemble is the most effective over all 11 attacks in terms of average defense success rate (prediction accuracy). (3) When defending adversarial example, the model verification ensemble defense is capable of repairing most of the adversarial perturbed input with high PSR. In particular, as shown in Table 6, DSR is further improved by TSR when the verification cannot agree with each other.

Improvement with Input-Model Ensemble. In Table 6, we also show the improvement of defense performance when we combine the output model verification layer with the input defense layer and build the input-model verification diversity ensemble. In our prototype, we apply a median filter with window size 2*2, a rotation with -12 degree, and a non-local mean filter NLM 13-3-4 to the CIFAR-10 input and then feed the processed input to multiple models. There can be other ways to combine the diverse input processing techniques with diverse output models and we consider it as future work. The input-model verification XEnsembles improves the model XEnsemble defense significantly.

Effect of Transferability. The transferability of adversarial examples is widely reported in literature [7]. An immediate